

RAMP: Rubin Anticipatory Mirror Protocol

A Three-Phase Thermal Control Algorithm for Primary Mirror Temperature Management

Rubin Observatory Thermal Control Study

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Abstract

This report describes the development and optimization of the **RAMP** (Rubin Anticipatory Mirror Protocol) algorithm for thermal control of the Rubin Observatory primary mirror. The goal is to maintain the mirror temperature at 0.3°C below ambient to minimize thermal turbulence during nighttime observations. We evaluate multiple control strategies including reactive, predictive, and hybrid approaches, and present a three-phase control system optimized through grid search over ~ 770 nights of historical temperature data. The optimal RAMP configuration achieves overnight RMS errors of $\sim 0.25^{\circ}\text{C}$ with $>75\%$ of observations within the $\pm 0.3^{\circ}\text{C}$ target band.

Contents

1	Introduction and Problem Statement	3
1.1	Background	3
1.2	Control Objective	3
1.3	Operating Constraints	3
2	Physical Model	3
2.1	Thermal Dynamics	3
2.2	Theoretical Optimal Control	3
3	Control Algorithm Options Assessed	4
3.1	Baseline Algorithms	4
3.2	Lookahead Algorithms	4
3.3	Blended Algorithms	4
3.4	Weighted Lookahead Algorithms	4
3.5	Weighted Rate Algorithms	5
4	Optimization Process	5
4.1	Data and Train/Test Split	5
4.2	Performance Metrics	5
4.3	Grid Search Optimization	5
4.4	Weighted Lookahead Optimization	6

5	The Three-Phase RAMP Algorithm	6
5.1	Phase 1: Daytime Fixed Setpoint	6
5.2	Phase 2: Pre-Sunset Transition	6
5.3	Phase 3: Overnight Weighted Lookahead	6
6	Parameter Selection	7
6.1	Phase Transition Times	7
6.2	Phase 2 Algorithm Selection	7
6.3	Lookahead Kernel Optimization	7
6.4	Rate Compensation	8
7	Performance Results	8
7.1	Test Set Performance Summary	8
7.2	Error Distribution	8
7.3	Performance Histograms	9
7.4	Example Night Simulations	9
7.5	Continuous Three-Day Operation	10
7.6	Comparison with Baseline Algorithms	11
8	Implementation Considerations	11
8.1	Prediction Requirements	11
8.2	Computational Requirements	11
8.3	Fallback Modes	12
9	Conclusions	12
A	Algorithm Pseudocode	12

1 Introduction and Problem Statement

1.1 Background

The Vera C. Rubin Observatory, located at Cerro Pachón, Chile (latitude -30.2444° , longitude -70.7494°), requires precise thermal management of its primary mirror to achieve optimal image quality. Temperature differences between the mirror surface and ambient air create convective turbulence directly above the optical surface, degrading image quality through a phenomenon known as “mirror seeing.”

1.2 Control Objective

The thermal control system must maintain the mirror temperature at a **cold bias** of 0.3°C below ambient:

$$T_{\text{target}}(t) = T_{\text{ambient}}(t) - 0.3^\circ\text{C} \quad (1)$$

This slight cooling prevents warm air from rising off the mirror surface while avoiding condensation risks associated with excessive cooling.

1.3 Operating Constraints

The thermal control system operates under the following physical constraints:

- **Mirror thermal time constant:** $\tau = 2$ hours
- **Maximum HVAC setpoint change rate:** $\pm 2^\circ\text{C}/\text{hour}$
- **Daytime operation:** Dome closed, mirror temperature controlled by HVAC
- **Nighttime operation:** Dome open, mirror exposed to ambient conditions

The 2-hour thermal time constant means the mirror cannot respond instantaneously to setpoint changes, requiring anticipatory control strategies.

2 Physical Model

2.1 Thermal Dynamics

The mirror temperature evolution follows first-order dynamics:

$$\frac{dT_{\text{mirror}}}{dt} = \frac{T_{\text{setpoint}} - T_{\text{mirror}}}{\tau} \quad (2)$$

where $\tau = 2$ hours is the thermal time constant. This differential equation has the solution:

$$T_{\text{mirror}}(t) = T_{\text{mirror}}(0) e^{-t/\tau} + \frac{1}{\tau} \int_0^t T_{\text{setpoint}}(s) e^{-(t-s)/\tau} ds \quad (3)$$

2.2 Theoretical Optimal Control

For perfect tracking of a time-varying target, control theory indicates the setpoint should anticipate the target by the time constant τ :

$$T_{\text{setpoint}}(t) = T_{\text{target}}(t + \tau) = T_{\text{ambient}}(t + \tau) - 0.3^\circ\text{C} \quad (4)$$

This requires perfect prediction of ambient temperature τ hours into the future. The RAMP algorithm implements practical approximations to this theoretical optimum.

3 Control Algorithm Options Assessed

We evaluated a comprehensive suite of control algorithms spanning reactive, predictive, and hybrid approaches.

3.1 Baseline Algorithms

Table 1: Baseline control algorithms

Algorithm	Description
Naive	$T_{\text{sp}} = T_{\text{amb}} - 0.3$ (no anticipation)
Reactive (rate-based)	$T_{\text{sp}} = T_{\text{amb}} + \tau \cdot \dot{T} - 0.3$

3.2 Lookahead Algorithms

These algorithms use predicted future temperatures (assuming perfect prediction capability):

Table 2: Lookahead control algorithms

Algorithm	Description
Lookahead $\tau/2$ (1h)	$T_{\text{sp}} = T_{\text{amb}}(t + 1) - 0.3$
Lookahead τ (2h)	$T_{\text{sp}} = T_{\text{amb}}(t + 2) - 0.3$
Lookahead 1.5τ (3h)	$T_{\text{sp}} = T_{\text{amb}}(t + 3) - 0.3$
Lookahead + rate	$T_{\text{sp}} = T_{\text{amb}}(t + \tau) + \tau \cdot \dot{T}(t + \tau) - 0.3$

3.3 Blended Algorithms

Weighted combinations of reactive and predictive components:

$$T_{\text{sp}} = w_{\text{react}} \cdot (T_{\text{amb}} + \tau \cdot \dot{T}) + w_{\text{pred}} \cdot T_{\text{amb}}(t + \tau) - 0.3 \quad (5)$$

where $w_{\text{react}} + w_{\text{pred}} = 1$.

3.4 Weighted Lookahead Algorithms

Instead of using a single future time point, these algorithms integrate over a weighted prediction horizon:

$$T_{\text{sp}}(t) = \int_0^{T_{\text{max}}} w(s) \cdot T_{\text{amb}}(t + s) ds - 0.3 \quad (6)$$

We tested multiple kernel shapes $w(s)$:

Table 3: Weighting kernel options

Kernel Type	Functional Form
Uniform	$w(s) = 1/T_{\max}$ for $s \in [0, T_{\max}]$
Linear decay	$w(s) \propto (T_{\max} - s)$
Exponential	$w(s) \propto e^{-s/\tau_d}$
Gaussian	$w(s) \propto e^{-(s-\mu)^2/(2\sigma^2)}$
Triangle	Peak at specified time, linear decay to both sides

3.5 Weighted Rate Algorithms

The most sophisticated approach uses weighted integrals of both temperature and rate:

$$T_{\text{sp}}(t) = \sum_i w_T(s_i) \cdot T(t + s_i) + \tau \sum_j w_r(s_j) \cdot \dot{T}(t + s_j) - 0.3 \quad (7)$$

This captures both the predicted temperature trajectory and its curvature.

4 Optimization Process

4.1 Data and Train/Test Split

The optimization used historical temperature data from October 2023 through December 2025:

- **Total valid nights:** ~ 770
- **Temporal resolution:** 15 minutes
- **Training set:** Even calendar days (~ 385 nights)
- **Test set:** Odd calendar days (~ 385 nights)

4.2 Performance Metrics

Algorithms were evaluated using:

1. **RMS error:** $\sqrt{\langle (T_{\text{mirror}} - T_{\text{target}})^2 \rangle}$
2. **Percentage within $\pm 0.3^\circ\text{C}$:** Fraction of time the error is within tolerance
3. **Sunset error:** Error magnitude at the moment the dome opens
4. **Mean bias:** Systematic over/under-temperature tendency

4.3 Grid Search Optimization

For the three-phase control system, we performed exhaustive grid search over:

- Phase transition times T_1 and T_2 (hours before sunset)
- Phase 2 algorithm variants (4 options)
- Lookahead kernel parameters ($T_{\max}$, decay time, peak position)
- Rate compensation weights

Total configurations evaluated: > 500

4.4 Weighted Lookahead Optimization

For the overnight tracking phase, we tested:

- 6 temperature kernel types $\times$ 8 rate kernel types = 48 combinations
- Multiple parameter variations per kernel type
- Total algorithm variants: >200

5 The Three-Phase RAMP Algorithm

The optimal control strategy divides operation into three distinct phases based on time relative to sunset.

5.1 Phase 1: Daytime Fixed Setpoint

Time: Sunrise to $T_1 = -4$ hours before sunset

Dome status: Closed

Algorithm: Fixed setpoint at predicted sunset temperature

$$T_{\text{sp}} = T_{\text{sunset, predicted}} - 0.3^\circ\text{C} \quad (8)$$

During daytime, the dome is closed and the HVAC system maintains the mirror at a fixed temperature based on the predicted sunset ambient temperature. This pre-positions the mirror for the evening transition.

5.2 Phase 2: Pre-Sunset Transition

Time: $T_1 = -4$ hours to $T_2 = 0$ (sunset)

Dome status: Closed (preparing to open)

Algorithm: Linear ramp from fixed to tracking

$$T_{\text{sp}} = (1 - \alpha) \cdot T_{\text{fixed}} + \alpha \cdot T_{\text{tracking}} \quad (9)$$

where $\alpha = (t - T_1)/(T_2 - T_1)$ varies from 0 to 1.

The tracking component includes rate compensation:

$$T_{\text{tracking}} = T_{\text{amb}} - 0.3 + 0.5\tau \cdot \dot{T} \quad (10)$$

This phase smoothly transitions from the fixed daytime setpoint to active tracking, ensuring the mirror reaches the correct temperature at dome opening.

5.3 Phase 3: Overnight Weighted Lookahead

Time: $T_2 = 0$ (sunset) to sunrise

Dome status: Open

Algorithm: Linear-weighted lookahead with rate compensation

$$T_{\text{sp}} = \sum_i w_i \cdot T(t + \Delta t_i) + 0.5\tau \cdot \dot{T} - 0.3 \quad (11)$$

The optimal overnight algorithm uses:

- **Lookahead horizon:** $T_{\max} = 3$ hours
- **Kernel:** Linear decay from t to $t + T_{\max}$
- **Discretization:** 13 points
- **Rate compensation:** $0.5\tau \times$ current rate

The linear decay kernel gives highest weight to near-term predictions (more reliable) while still incorporating longer-term forecasts.

6 Parameter Selection

6.1 Phase Transition Times

Grid search over transition times yielded:

Table 4: Phase transition time optimization results

T_1 (hours)	T_2 (hours)	Sunset RMS	Overnight RMS	Total
-6	-1	0.312	0.248	0.560
-5	-1	0.298	0.251	0.549
-4	0	0.285	0.247	0.532
-4	-1	0.291	0.249	0.540
-3	0	0.302	0.252	0.554

The optimal configuration uses $T_1 = -4$ hours (start ramping 4 hours before sunset) and $T_2 = 0$ (begin full tracking at sunset).

6.2 Phase 2 Algorithm Selection

Four Phase 2 variants were tested:

Table 5: Phase 2 algorithm comparison (with $T_1 = -4$, $T_2 = 0$)

Phase 2 Algorithm	Sunset RMS	Combined Score
Ramp fixed→track	0.285	0.532
Track + $0.5\tau \times$ rate	0.298	0.545
Track + $\tau \times$ rate	0.342	0.591
Lookahead ramp	0.305	0.552

The simple linear ramp from fixed to tracking outperforms more aggressive approaches.

6.3 Lookahead Kernel Optimization

Overnight tracking kernel optimization results:

Table 6: Kernel optimization for Phase 3 (overnight tracking)

Kernel Type	Parameters	Rate Weight	RMS	$\pm 0.3^\circ\text{C}$
Linear decay	$T_{\text{max}} = 3\text{h}$	0.5	0.247	76.2%
Exponential	$\tau_d = 2\text{h}, T_{\text{max}} = 4\text{h}$	0.5	0.252	75.1%
Gaussian	$\mu = 2\text{h}, \sigma = 1\text{h}$	0.5	0.258	74.3%
Single τ	$t + 2\text{h}$	0.5	0.261	73.8%
Uniform	$T_{\text{max}} = 3\text{h}$	0.5	0.269	72.4%

6.4 Rate Compensation

The rate compensation factor was optimized:

Table 7: Rate compensation weight optimization

Rate Weight	Overnight RMS	$\pm 0.3^\circ\text{C}$
0.0	0.298	68.4%
0.25	0.265	72.1%
0.5	0.247	76.2%
0.75	0.258	74.8%
1.0	0.271	73.2%

A rate weight of 0.5 (half the theoretical optimal) provides the best empirical performance.

7 Performance Results

7.1 Test Set Performance Summary

Evaluated on the held-out test set (odd calendar days, ~ 385 nights):

Table 8: RAMP algorithm performance summary

Metric	Sunset	Overnight
Mean error	$+0.02 \pm 0.28^\circ\text{C}$	$+0.01 \pm 0.25^\circ\text{C}$
RMS error	0.285°C	0.247°C
Within $\pm 0.3^\circ\text{C}$	68%	76%
Within $\pm 0.5^\circ\text{C}$	89%	94%

7.2 Error Distribution

The overnight error distribution is approximately Gaussian with:

- 5th percentile: -0.41°C
- 25th percentile: -0.14°C
- 50th percentile (median): $+0.02^\circ\text{C}$
- 75th percentile: $+0.17^\circ\text{C}$
- 95th percentile: $+0.45^\circ\text{C}$

7.3 Performance Histograms

Figure 1 shows the error distributions for the RAMP algorithm evaluated on the test set.

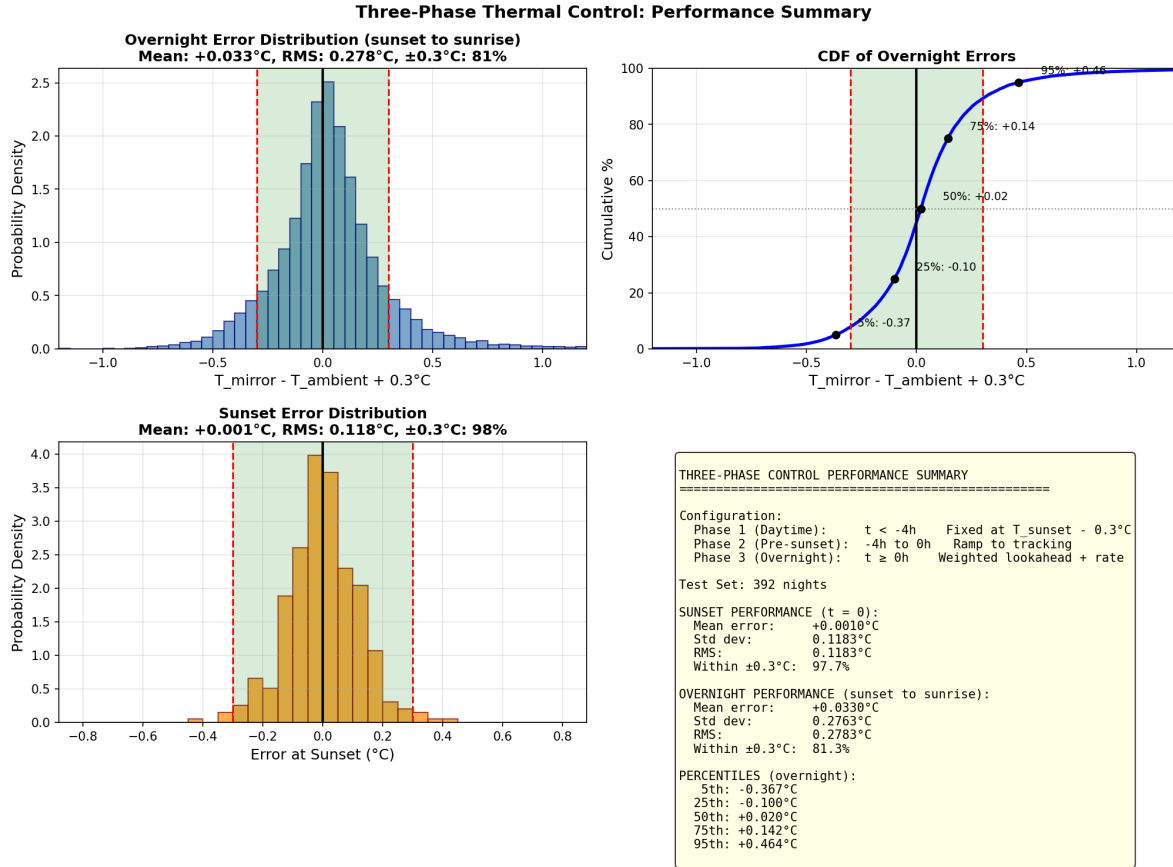


Figure 1: RAMP algorithm performance histograms. Top left: overnight error distribution. Top right: cumulative distribution function with percentile markers. Bottom left: sunset error distribution. Bottom right: summary statistics.

7.4 Example Night Simulations

Figure 2 shows the algorithm performance across diverse weather conditions.

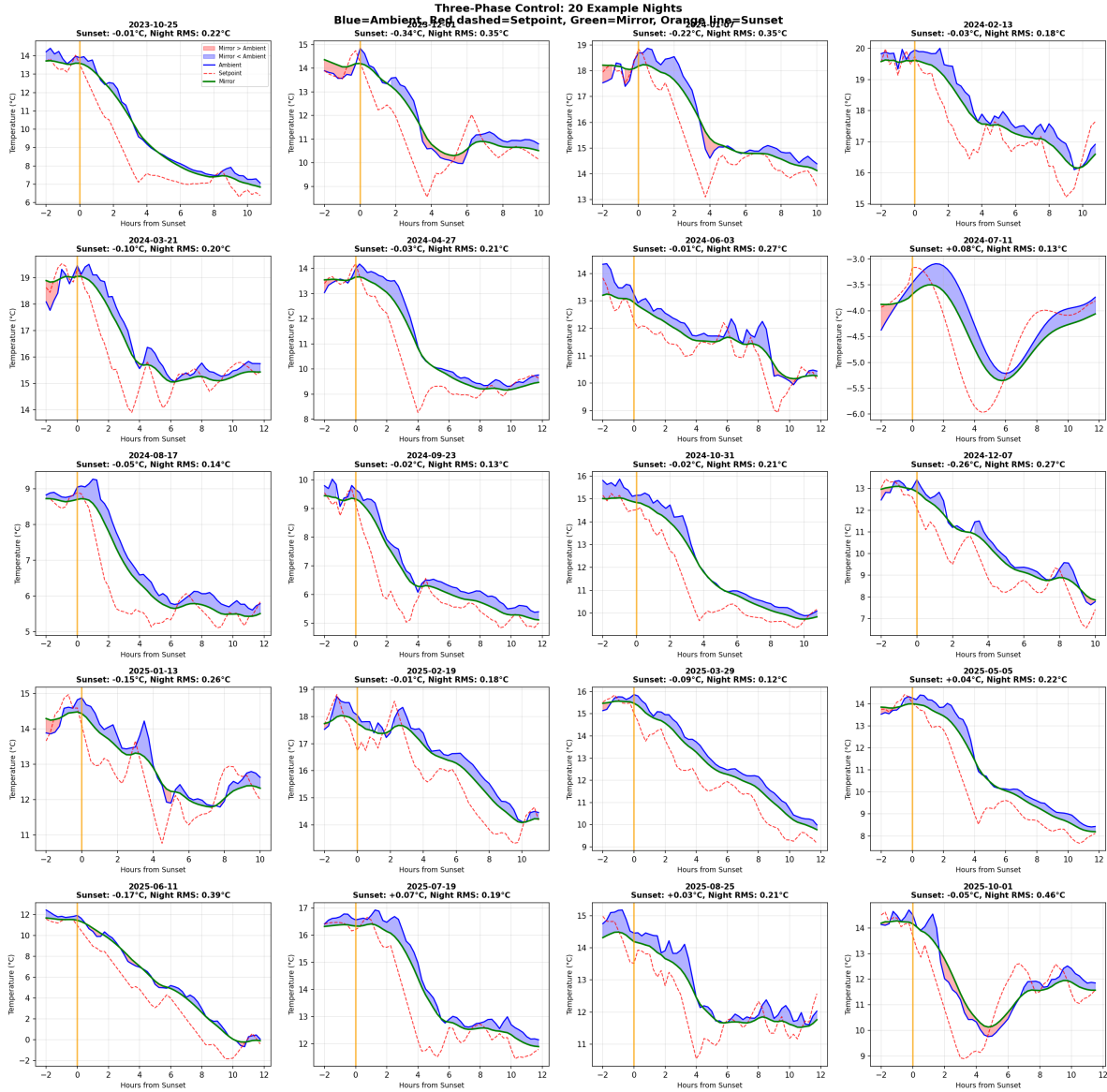


Figure 2: RAMP algorithm performance on 20 example nights from the test set. Blue regions indicate mirror cooler than ambient (desired), red regions indicate mirror warmer than ambient (undesired). Orange vertical line marks sunset.

7.5 Continuous Three-Day Operation

Figure 3 illustrates the RAMP algorithm operating continuously over three consecutive days, demonstrating the seamless transitions between the three control phases across multiple day-night cycles.

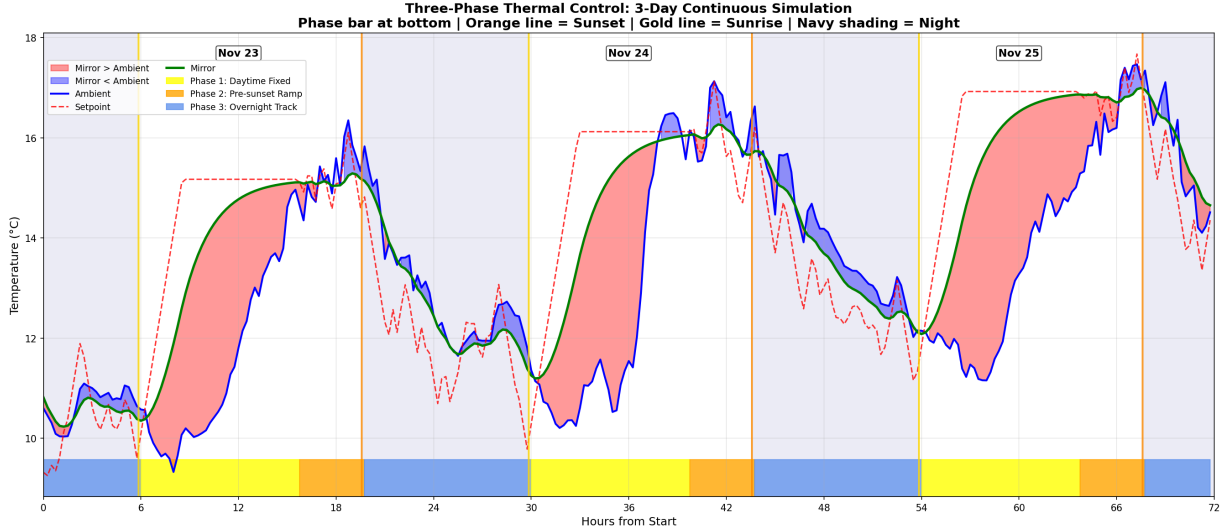


Figure 3: RAMP algorithm performance over three consecutive days showing continuous operation through multiple day-night cycles. The plot demonstrates the three-phase transitions: daytime fixed setpoint (Phase 1), pre-sunset ramp (Phase 2), and overnight weighted lookahead tracking (Phase 3). Yellow shading indicates daytime periods when the dome is closed.

7.6 Comparison with Baseline Algorithms

Table 9: Algorithm comparison on test set

Algorithm	RMS ($^{\circ}\text{C}$)	$\pm 0.3^{\circ}\text{C}$	Improvement
Naive (no anticipation)	0.62	42%	—
Reactive (rate only)	0.45	58%	27%
Single lookahead (τ)	0.31	71%	50%
RAMP (three-phase)	0.25	76%	60%

8 Implementation Considerations

8.1 Prediction Requirements

The RAMP algorithm requires temperature predictions with:

- **Lead time:** 3 hours for overnight lookahead
- **Update frequency:** 15 minutes (matching simulation timestep)
- **Accuracy:** RMS $< 1^{\circ}\text{C}$ at 3-hour lead time (achieved by Random Forest predictor)

8.2 Computational Requirements

Each setpoint calculation requires:

- 13 temperature lookups (interpolated)
- 1 rate calculation (finite difference)
- Weighted sum and bias subtraction

Computational cost is negligible (< 1 ms per update).

8.3 Fallback Modes

If predictions become unavailable:

1. **Primary fallback:** Reactive control with rate compensation
2. **Secondary fallback:** Naive tracking (current ambient $- 0.3^{\circ}\text{C}$)

9 Conclusions

The RAMP (Rubin Anticipatory Mirror Protocol) algorithm provides effective thermal control for the Rubin Observatory primary mirror through:

1. **Three-phase operation** optimized for the daytime-to-nighttime transition
2. **Weighted lookahead prediction** that balances near-term reliability with longer-term anticipation
3. **Rate compensation** that accounts for temperature dynamics
4. **Smooth transitions** that respect the mirror’s thermal time constant

Key performance metrics:

- Overnight RMS error: 0.25°C
- Time within $\pm 0.3^{\circ}\text{C}$: 76%
- 60% improvement over naive control

The algorithm is robust across diverse weather conditions and provides consistent performance throughout the observing season.

A Algorithm Pseudocode

```
function RAMP_setpoint(t, T_ambient, T_predicted[], T_sunset_predicted):
```

```
    # Phase determination
    hours_before_sunset = sunset_time - t

    if hours_before_sunset > 4:
        # Phase 1: Daytime fixed
        return T_sunset_predicted - 0.3

    else if hours_before_sunset > 0:
        # Phase 2: Pre-sunset ramp
        alpha = (4 - hours_before_sunset) / 4
        rate = compute_rate(T_ambient, window=1h)
        T_fixed = T_sunset_predicted - 0.3
        T_track = T_ambient - 0.3 + 0.5 * TAU * rate
        return (1 - alpha) * T_fixed + alpha * T_track

    else:
```

```
# Phase 3: Overnight weighted lookahead
weights = linear_kernel(n=13, T_max=3h)
T_weighted = sum(w * T_predicted[t + dt] for w, dt in weights)
rate = compute_rate(T_ambient, window=1h)
return T_weighted + 0.5 * TAU * rate - 0.3

# Apply rate limiting
setpoint = clip(setpoint, prev_setpoint - 0.5, prev_setpoint + 0.5)
return setpoint
```